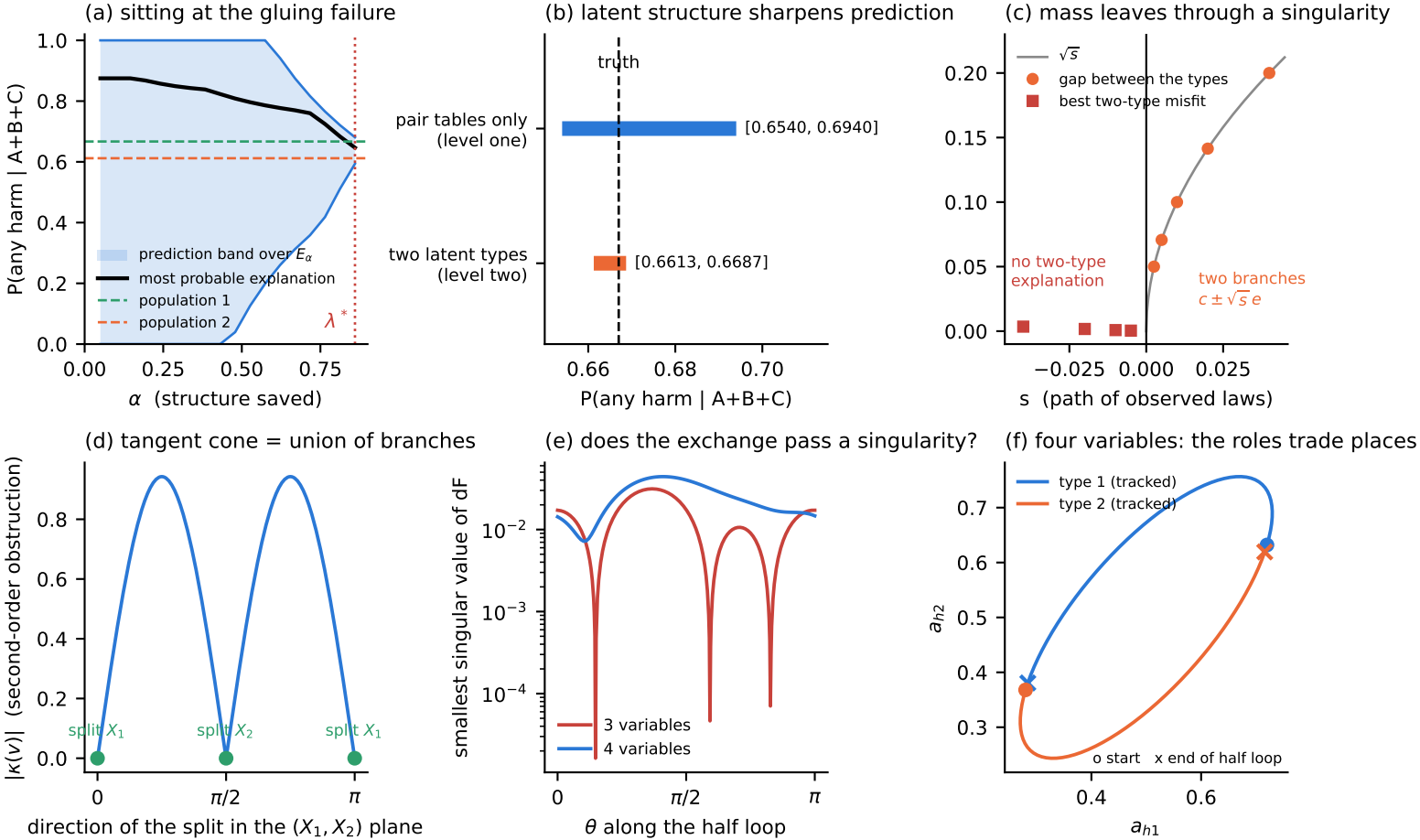
(a) sitting at the gluing failure
P(any harm | A+B+C)
1.0
0.8
0.6
0.4
0.2
0.0
0.00
0.25
0.50
0.75
α (structure saved)
prediction band over E_α
most probable explanation
population 1
population 2
λ^*
(b) latent structure sharpens prediction
truth
pair tables only
(level one)
[0.6540, 0.6940]
two latent types
(level two)
[0.6613, 0.6687]
0.66
0.68
0.70
P(any harm | A+B+C)
(c) mass leaves through a singularity
$\sqrt{s}$
gap between the types
best two-type misfit
0.20
0.15
0.10
0.05
0.00
no two-type
explanation
two branches
$c \pm \sqrt{s}\, e$
−0.025
0.000
0.025
s (path of observed laws)
(d) tangent cone = union of branches
|K(v)| (second-order obstruction)
0.8
0.6
0.4
0.2
0.0
split X_1
split X_2
split X_1
0
π/2
π
direction of the split in the (X_1, X_2) plane
(e) does the exchange pass a singularity?
smallest singular value of dF
10^{-2}
10^{-3}
10^{-4}
3 variables
4 variables
0
π/2
π
θ along the half loop
(f) four variables: the roles trade places
type 1 (tracked)
type 2 (tracked)
a_{h2}
0.7
0.6
0.5
0.4
0.3
o start
x end of half loop
0.4
0.6
a_{h1}